Middle East Technical University Department of Computer Engineering

CENG 351 Data Management & File Structures

Fall 2025–2026

In-Class Assignment 1 — ER

Date: 22 October 2025, Duration: 100 minutes

Student ID: ANSWER KEY Name and Surname: ANSWER KEY

(Also, write your name, surname, and Student ID on the solution page.)

1 Background

Imagine you are building a ModelHub platform like Hugging Face, where machine learning models are released by organizations, accounts have profiles, users upload datasets, users follow each other, users run specific model versions on datasets to create results, and results may be included in publications. The system should track **accounts (users and organizations) with profiles, models and their versions, datasets, results, and publications.** Your task is to (a) model this platform with an ER diagram and (b) list the schemata of the tables, adhering to the specifications below.

2 Specifications

Accounts in this system are either users or organizations. For accounts, information such as "account_name", "email", "joined_date", "status" (e.g., active, suspended, banned), and "AccountID" should be stored. Note that "AccountID" uniquely defines an account worldwide. Users have additional information to store their "reputation_score", which indicates a non-negative numeric score representing community trust and contribution quality. For organizations, "rating" information should be kept; it represents the average of ratings on a 0 to 5 scale for organizations. Users can follow other users, but organizations cannot follow anyone. A user can follow zero or more other users and can be followed by zero or more users. Each follow must be recorded with the follower, followee, and the corresponding "following_date" to allow querying of a user's followers and followings. Since a user can follow another only once, each follower–followee pair must be stored only once in the database.

Profiles are uniquely identified by "ProfileID"; also "bio" and "avatar_url" information should be stored. Each profile must belong to exactly one account, and each account may have at most one profile, since some accounts may choose not to create a profile page on the platform.

Models are recorded with "ModelID" (each model is identified by a unique "ModelID"), "model_name", "license", and "size" (interpreted as the model's number of parameters). An organization may release zero or more models; every model must be released by exactly one organization.

A model has zero or more model versions. Each *ModelVersion* is identified by a "version_no" that is unique within its model (the same "version_no" may appear in different models) and cannot exist without its model. The "version_date" for each model version is also recorded.

Datasets are recorded with "DatasetID" (each dataset is identified by a unique "DatasetID"), "dataset_name", "modality" (e.g., text, code, image, speech), and "number_of_rows" (the number of records in the dataset). A user may upload zero or more datasets, and each dataset must be uploaded by at least one user. Multiple users can upload the same dataset. For each upload of a dataset by a user, record the user's "role" for this upload (e.g., owner, contributor, validator, annotator). A user runs a specific model version on a dataset. In each run, one model version is executed on one dataset by a single user. Any user, model version, or dataset may participate in zero or more runs. A user may run multiple model versions on multiple datasets, or none. A dataset may be used in runs by multiple users with multiple model versions, or not at all. Similarly, a model version may be run by multiple users on multiple datasets, or not at all. For each run, record "run_date" and "run_type" (e.g., finetuned, zero-shot, few-shot, pretrained, evaluation). For the same user, model version, and dataset combination, keep a single run; if the user executes again with the same combination, the existing run must be overwritten so that the "run_date" and "run_type" reflect the latest execution.

Results are stored with a unique "ResultID" and include key information such as "accuracy" and "f1_score" (both evaluation metrics measured in the range 0 to 1). Each **run** may create zero or more results because it can evaluate multiple "hyperparameter_config" values; different hyperparameter configurations (such as learning_rate, batch_size, optimizer) create different results. Some runs may create no results (e.g., aborted or failed). Every result must be created by exactly one successful run. For each created result, record its "hyperparameter_config".

Publications are recorded with "PubID" (each publication is identified by a unique "PubID"), "title", "venue", "publication_year". Each publication recorded on the platform must include one or more results. A result may appear in zero or more publications, as some may remain unpublished. Each included result should be recorded with a "placement_type" (e.g., figure, table, text) and a "placement_section" (e.g., Experiments, Results, Related Work).

3 Output

a Draw the ER diagram corresponding to the above specifications. (50 pts.)

b List the schemata of the tables for the conceptual design you made. (50 pts.)
E.g.: Employees (<u>ssn</u>, name, lot)
Dept_Mgr (<u>did</u>, dname, budget, ssn, since) ssn REFERENCES Employees (ssn)

Note: Try to use as small number of tables as possible. Do not forget to underline the primary keys (both in ER and schemata) of the tables and show the foreign keys (in schemata). You don't need to write SQL Create Table commands.

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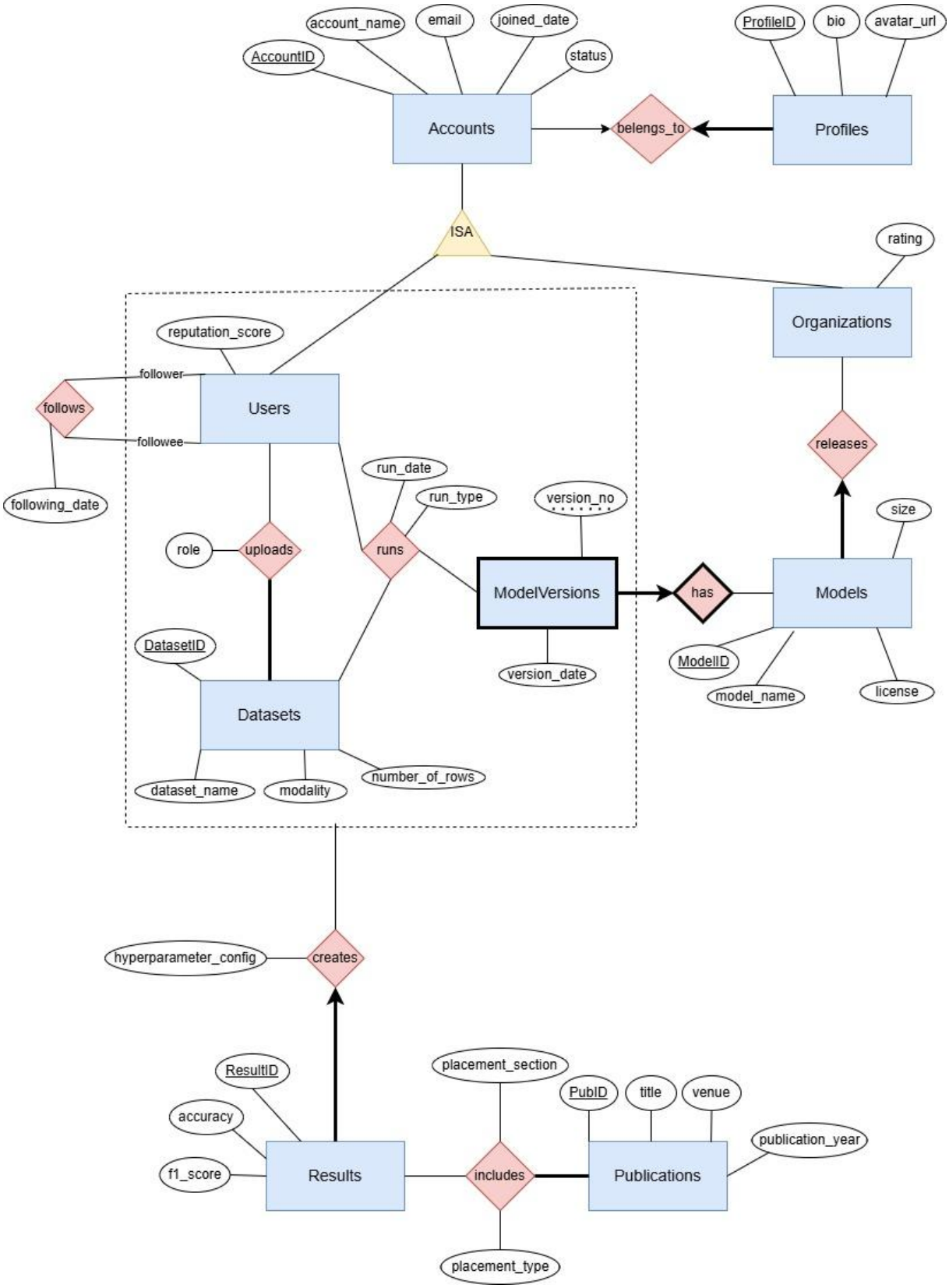
Solution

a) ER diagram solution is the following:



b) List of schemata for the solution is the following:

Accounts (AccountID, account_name, email, joined_date, status)

Users (AccountID, reputation_score) AccountID REFERENCES Accounts (AccountID)

Organizations (AccountID, rating) AccountID REFERENCES Accounts (AccountID)

Profiles (ProfileID, bio, avatar_url, AccountID) AccountID REFERENCES Accounts (AccountID)

Note: AccountID NOT NULL and UNIQUE

follows (follower_id, followee_id, following_date) follower_id REFERENCES Users (AccountID), followee_id REFERENCES Users (AccountID)

Models (ModelID, model_name, license, size, AccountID) AccountID REFERENCES Organizations (AccountID)

Note: AccountID NOT NULL

ModelVersions (ModelID, version_no, version_date) ModelID REFERENCES Models (ModelID)

Datasets (DatasetID, dataset_name, modality, number_of_rows)

uploads (AccountID, DatasetID, role) AccountID REFERENCES Users (AccountID), DatasetID REFERENCES Datasets (DatasetID)

runs (AccountID, ModelID, version_no, DatasetID, run_date, run_type) AccountID REFERENCES Users (AccountID), (ModelID, version_no) REFERENCES ModelVersions (ModelID, version_no), DatasetID REFERENCES Datasets (DatasetID)

Results (ResultID, accuracy, f1_score, hyperparameter_config, AccountID, ModelID, version_no, DatasetID) (AccountID, ModelID, version_no, DatasetID) REFERENCES runs (AccountID, ModelID, version_no, DatasetID)

Note: (AccountID, ModelID, version_no, DatasetID) NOT NULL

Publications (PubID, title, venue, publication_year)

includes (PubID, ResultID, placement_type, placement_section) PubID REFERENCES Publications (PubID), ResultID REFERENCES Results (ResultID)